# Tension-compression asymmetry of metastable austenitic stainless steel studied by in-situ high-energy X-ray diffraction

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**Abstract**
This work studies the tension-compression asymmetry (TCA) of metastable austenitic stainless steel (MASS) in uniaxial loading depending on temperature. In-situ high-energy X-ray diffraction (HEXRD) was used to simultaneously probe phase fractions, transformation kinetics, crystallographic texture, lattice strains, strain and stress partitioning between austenite and martensites during quasistatic tensile and compressive deformation at 24°C, 60°C and 100°C. Complementary relaxed-constraint crystal plasticity simulations and calculations of the mechanical driving force related to the formation of α' and ε martensites were performed. At 24°C, martensitic transformations (MTs) prevail while at 100°C dislocation slip is the dominant deformation mechanism for both load senses. Macroscopic stress-strain response and transformation behaviour exhibit weak TCA, with compression promoting the conversion of ε into α'. Transformation kinetics were analysed in relation to shear banding and the geometric alignment of ε lamellas depending on load sense and temperature. A strong TCA was found for crystallographic texture, bearing signatures of grain rotation due to plastic slip and of MT in case of austenite (γ). For both load senses, the relative strengths of austenite and martensite texture fibres were related to the driving force anisotropy for α' formation calculated based on the phenomenological theory of martensite crystallography. Texture evolution of α' is largely controlled by the MT itself, not by grain rotation. Analysis of differently oriented austenite grain families revealed a pronounced TCA of the lattice strains, linked to the γ → ε MT. This was found to be a direct consequence of driving force and volume change related to ε formation. Furthermore, stress is shared differently between austenite and martensites in tension vs. in compression. γ hardens more and hence carries a larger portion of the total stress in compression than in tension. These findings can help aid the development of new metastable austenitic stainless steels (MASS) material laws that are sensitive to load-sense and temperature for advanced forming simulations.

Keywords: Transformation-induced plasticity, TRIP, micromechanics, driving force, texture, volume change

## 1. Introduction

Metastable austenitic stainless steels (MASS) are a special class of transformation induced plasticity (TRIP) steels known for their unique combination of strength, formability, and corrosion resistance [1]. They are indispensable materials in the transport sector, chemical industries, power generation, automotive and aerospace, home appliances and food industry, making up 75% of the world production of stainless steel [2, 3]. The high deformability of MASS allows them to be designed as sheet products that can be shaped into strong and complex parts, enabling optimized geometries and weight savings. Benefitting technologies include bipolar plates for fuel cells [4] and electric vehicle battery housings [5], structural components which are critical for the green transition.

While the ‘stainlessness’ results from a Cr content of at least about 14 wt.%, the metastable nature of the austenite phase bestows superior formability and toughness to MASS. Underlying are mechanically induced phase changes, where the fcc austenite, denoted γ, transforms into one or two martensitic phases during deformation. Both the γ → α’ and the γ → ε (→ α’) pathways have been reported [6-11], where austenite either transforms into bcc α’ or into hcp ε, while ε often serves as a transitional phase, which further converts into α’ as deformation progresses. The content of martensite and its distribution in the final part geometry have direct consequences for manufacture and in-service performance of MASS, including: springback prediction, which is particularly challenging for TRIP steels [12-14]; delayed cracking of deep-drawn Cr-Ni stainless steel [15]; resistance to high-cycle and very high-cycle fatigue [16].

Previous studies indicate that the transformation behaviour is influenced by many factors, including external conditions such as temperature [7, 8, 17-19], strain mode [19-22], strain rate [20, 21, 23], next to material characteristics, such as composition [8, 18, 19, 24, 25], initial grain size [6, 11, 22, 26], grain orientation and crystallographic texture [11, 27, 28].

Due to its tremendous impact on the mechanical stability of austenite, temperature has the strongest effect on transformation and hardening of any MASS. With decreasing temperature, the sigmoidal stress-strain behaviour of unstable austenite (e.g. EN 1.4318 at room temperature (RT) or EN 1.4401 at low temperature) replaces the parabolic flow curve of highly stabilized austenite (e.g. EN 1.4401 at RT or EN 1.4318 at high temperature) [8, 29]. The influence of temperature is such that the mechanical energy dissipated as heat and the latent heat locally released during martensitic transformation (MT) can stabilize the austenite and thus inhibit TRIP [6, 23]. This situation becomes exacerbated by the low thermal conductivity of austenitic stainless steel (~16 $Wm^{-1}K^{-1}$ for the grade studied here), causing local temperature variations at transformation hot spots, which affect the amount of martensite formed.

During forming and service life MASS undergoes various stress modes and in general non-monotonic strain paths. Predicting the influence of stress modes and strain paths on TRIP and on component behaviour is complex and challenging, because martensite content, texture and internal (residual) stresses are commonly affected by mechanical loading conditions and their evolution. For instance, Murr et al. [21] reported for a 304 type MASS faster transformation kinetics and more blocky α’ martensite in biaxial tension than in uniaxial tension. Varma et al. [22] found that rolling produces considerably more martensite than uniaxial tension for 304 and 316 grade MASS.

Drawing general conclusions is unfortunately difficult because of conflicting findings reported in the literature. Lebedev and Kosarchuk [30] studied two kinds of 18-10 Cr-Ni MASS and found faster kinetics for both α’ and ε martensites for deformation by tension than by torsion and especially by compression. Iwamoto et al. [31] on the other hand measured a larger martensite content for compression than for tension at low strains, and the inverse in the high-strain region, for 304 type MASS. However, phase transformations are not the only possible source for mechanical anisotropy [32]. For instance, residual stresses [33, 34], crystallographic texture [34], and kinematic hardening due to pre-deformation [14] on their own are known to lead to pronounced anisotropic behaviour in MASS, for instance in the form of a tension-compression asymmetry (TCA), adding further complexity.

However, to optimize shaping operations and component performance, industrial simulation tools capable of predicting the impact of different stress states and strain paths are becoming increasingly vital [35]. These tools require accurate constitutive laws linking material behaviour to imposed deformation, allowing to solve the governing elastoplastic continuum mechanics equations, which is usually achieved numerically by finite element methods. The material laws employed are increasingly multi-level, involving features on multiple length-scales [36-38]. Calibration and validation of these laws, whether phenomenological or more physics-based, need detailed reference data, ideally from in-situ experiments probing multiple scales simultaneously.

In-situ high-energy X-ray diffraction (HEXRD) is a powerful tool for this purpose, as it provides simultaneous access to phase fractions and transformation kinetics, crystallographic texture, lattice strains, and sharing of stress between phases [39-43]. The high-flux high-energy X-ray beam allows penetrating mm thick steel specimens and fast sampling of the bulk material state with (detector exposure) rates of 10 Hz and more, easily sufficient for quasi-static tests. By linking in-situ recorded diffraction patterns to stress-strain data from the load rig, microstructure, micromechanics and macroscopic behaviour can be connected, aiding the development of advanced material constitutive laws for TRIP steels [36, 37, 44]. However, most in-situ HEXRD studies related to the mechanical behaviour of MASS are limited to uniaxial tension [36, 40-42, 44], while only very few works investigate other strain modes [37, 39].

The main goal of this work is to study the influence of the uniaxial load sense and of temperature on the mechanical behaviour and TRIP-related micromechanics of EN 1.4318 (AISI 301LN) MASS. This steel exhibits low austenite stability and thus strong propensity to form strain-induced martensite at room temperature. We use in-situ HEXRD to simultaneously probe transformation behaviour, texture evolution and partitioning of stress during tensile and compressive loading at 24°C, 60°C, and 100°C.

The first objective is to correlate macroscopic flow and work-hardening with the transformation behaviour. Phase fractions of martensites α' and ε and their transformation kinetics are analysed, considering differences in the transformation pathway and shear banding between tension and compression. Specific reference is made to the four stage work-hardening concept introduced by Talonen [45], which applies well at 24°C for both load senses but needs to be adapted for elevated temperature.

The second objective is to reveal the deformation processes responsible for the strong texture TCA, which we observe at all temperatures. Mechanisms taken into account include grain rotation due to dislocation slip, denuding of austenite texture fibres due to MT, and the transformation texture of α' martensite. To provide the transformation-free baseline for texture evolution depending on load sense, we employ crystal plasticity (CP) simulations with the ALAMEL relaxed constraint model [46]. In order to account for variant selection of strain-induced martensite, the mechanical driving force for $\gamma \rightarrow \alpha'$ is examined and its anisotropy related to the texture measurements.

The third objective is to study the influence of load sense and temperature on the partitioning of strain and stress (i) among differently oriented families of austenite and martensite grains and (ii) between austenite and martensites. Particular attention is paid to the impact of the volume changes linked to the MTs and the directionality of the mechanical driving force of $\gamma \rightarrow \varepsilon$ to explain the pronounced TCA of austenite lattice strains. Eventually, the phase-specific contributions of austenite and martensites to the total macroscopic stress are estimated.

## 2. Experimental and computational procedures

### 2.1. Material preparation and characterization

The material studied is EN 1.4318 (AISI 301LN) MASS with the composition specified in Table 1. The C content was measured coulometrically (Coulomat 702 SO/CS), O, N and H contents via fusion analysis, while other element concentrations were determined by inductively coupled plasma optical emission spectroscopy (ICP-OES, Varian 720 ES).

Samples for light-optical microscopy were prepared by grinding, diamond polishing and electro-etching in 65% $HNO_3$. Texture in the as-received state was measured by XRD (Bruker D8 Advance, Cu Kα radiation) using an energy-dispersive compound silicon strip detector for filtering sample fluorescence (LynxEye XE-T).

The rolled, as-received material exhibited a microstructure consisting of austenite (face centred cubic - fcc) with a grain size of ~ 30 µm and a mostly random crystallographic texture, as the {111} and {200} pole figures in Figure 1 illustrate. Austenite grain boundaries oriented parallel to the plate plane were sporadically decorated by broken bands of δ ferrite, originating from the steel's solidification mode [47]. Their volume fraction was (3.3 ± 1.4) % as measured by image segmentation. Comparable δ contents are commonly observed in similar steels [6, 47-49].

Table 1: EN 1.4318 (AISI 301LN) steel composition in wt.%.

| | Fe | Cr | Ni | Mn | Si | Mo | Cu | N | Co | P | S | C | Ti | O | H |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Average | 75.3 | 16.9 | 5.7 | 1.07 | 0.67 | 0.144 | 0.096 | 0.096 | 0.065 | 0.0288 | 0.020 | 0.0093 | 0.007 | 0.002 | 0.001 |
| Standard deviation | 0.4 | 0.2 | 0.1 | 0.01 | 0.05 | 0.002 | 0.001 | | 0.001 | 0.0001 | 0.001 | 0.0004 | 0.001 | | |

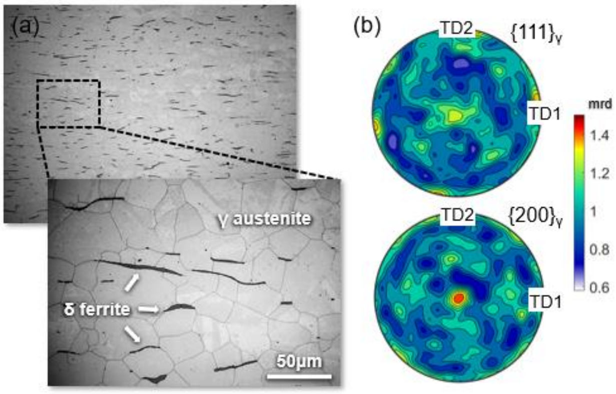


Figure 1: As-received microstructure. (a) EBSD-IPF map and (b) recalculated XRD pole figures. The sample reference frame used in (b) is defined in Figure 2.

The deformation behaviour of austenitic stainless steel is crucially influenced by the energy required to form an intrinsic stacking fault on $\{111\}_\gamma$. A lower stacking fault energy (SFE) leads to more widely spread partial dislocations, which move readily away from each other under the influence of an external stress [50]. In MASS this results in the formation of planar obstacles including individual stacking faults, thin ε martensite lamellae and in other cases deformation twins [7, 9, 18, 27]. Twins and ε martensite arise if stacking faults overlap on adjacent and on alternating {111} planes, respectively. These extended planar faults cannot be easily overcome by cross-slip and therefore strongly increase the material's ability to work hardening. In general, a lower SFE has been found to promote martensite formation over twinning and conventional dislocation slip. For SFEs larger than 30−40 mJ/m$^2$ conventional slip and stacking fault creation dominate over martensite formation, while for SFEs lower than 18−20 mJ/m$^2$ martensite formation dominates [29]. Table 1 compares the SFE of the steel studied here with two industrially widely used MASS standards according to empirical relations reported in the literature [51-54]. The EN 1.4318 (AISI 301LN) steel of the present work exhibits a SFE < 20 mJ/m$^2$. Despite the differences in the SFE estimates depending on literature source, the SFE of EN 1.4318 (AISI 301LN) is invariably smaller than that of EN 1.4301 (AISI 304) and of EN 1.4401 (AISI 316). Also, the steel's $M_s$ temperatures are close to RT, as Table 3 shows.

The relative importance of the SFE vs. the austenite-martensite Gibbs free energy difference for temperature and composition sensitivities of TRIP is disputed [1, 9]. The SFE in MASS grows weakly on heating from RT and 100°C, with reported increases ranging between 3.8 and 7.6 mJ/m$^2$ for Cr-Ni steels similar to the formulation studied here [18, 55-57]. However, the details of the influence of alloying elements on the SFE are complex and necessitate further close investigation [58].

Table 2: Comparison of stacking fault energies (SFE) of metastable austenitic stainless steels based on composition dependencies reported in literature (all values in mJ/m$^2$).

| Steel | Schramm, Reed (1975) [51] | Rhodes, Thompson (1977) [52] | Brofman, Ansell (1978) [53] | Dai et al. (2002) [54] |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| EN 1.4318 (AISI 301LN) [this work, composition according to Table 1] | -0.9 | 8.5 | 13.8 | ~17 |
| EN 1.4301 (AISI 304) [59] | 9−32 | 23−30 | 18−23 | 21−31 |
| EN 1.4401 (AISI 316) [59] | 39−81 | 25−34 | 23−32 | 58−80 |

Table 3: Comparison of start temperatures for α' martensite ($M_{s,\alpha'}$) and ε martensite ($M_{s,\varepsilon}$) of metastable austenitic stainless steels based on composition dependencies reported in literature (all values in °C).

| | $M_{s,\alpha'}$ | | $M_{s,\varepsilon}$ |
|---|---|---|---|
| Steel | Dai et al. (2002) [54] | Hammond (1964) [60] | Dai et al. (2002) [54] |
| EN 1.4318 (AISI 301LN) [this work, composition according to Table 1] | 32 | 17 | 29 |
| EN 1.4301 (AISI 304) [59] | -180−70 | -520−90 | -140−60 |
| EN 1.4401 (AISI 316) [59] | -440−90 | -800− -50 | -200− -20 |

## 2.2. Real time in-situ high-energy XRD

Samples for in-situ HEXRD loading were extracted by electric discharge machining along the sheet transverse direction. Specimens for tensile and compressive loading were flat dog-bones and round cylinders, respectively (Figure 2). HEXRD was performed at beamline P07 of DESY (Germany) using a monochromatic X-ray beam of 100 keV. Samples were loaded at three different temperatures (24°C, 60°C, 100°C) in tension and compression at a strain rate of $1\cdot10^{-3}$/s using the modified Bähr deformation dilatometer available at the beamline. 10 kN and 20 kN load cells were used for tension and compression, respectively. Two-dimensional (2D) transmission diffraction patterns were recorded on a Perkin Elmer image plate detector (2048·2048 pixels, exposure time 1 s) while continuously deforming the samples. For all samples the diffracting volume was larger than 1.5 $mm^3$ at an X-ray spot size of 0.8·0.8 $mm^2$. The sample-to-detector distance, beam centre on the detector, detector tilt and instrumental broadening were calibrated with $LaB_6$ standard powder. The dilatometer load direction was parallel to the horizontal detector axis (Figure 2b).

The full 2D diffraction patterns were evaluated by combined Rietveld analyses to extract phase-specific information including volume fraction, crystallographic texture, phase- and lattice plane dependent microstrain depending on the imposed deformation. Each pattern was azimuthally segmented (along η in Figure 2) into 72 wedges, 5° each, which were then refined simultaneously using Maud [61, 62]. The extended Williams-Imhof-Matthiesen-Vinel (E-WIMV) algorithm [63, 64] was employed to account for preferred orientation of the crystallites in the diffracting volume. For microstrain and phase fractions, refinements were carried out with arbitrary texture to permit best possible fit of diffracted intensities.

Uniaxial stress tests are characterized by a superposition of hydrostatic and deviatoric stress of the specific form

$$\sigma = \sigma_h + \sigma_d = \begin{bmatrix} s_h & 0 & 0 \\ 0 & s_h & 0 \\ 0 & 0 & s_h \end{bmatrix} + \begin{bmatrix} 2s_h & 0 & 0 \\ 0 & -s_h & 0 \\ 0 & 0 & -s_h \end{bmatrix} \quad (1)$$

where $s_h = s/3$ is the hydrostatic stress and $s$ is the stress applied along the load axis. Importantly, this stress state is a special case of that found in a diamond anvil cell (DAC). Yet, while in the DAC the total stress is triaxial and the hydrostatic component strongly dominates, for tension and compression tests as those performed here the transverse normal total stresses (components 22 and 33 of $\sigma$ in Equation 1) are zero. Lattice plane-dependent microstrain (type I internal strain) in uniaxial tests are therefore appropriately described via the DAC strain model implementation in Maud [61, 65], as carried out here.

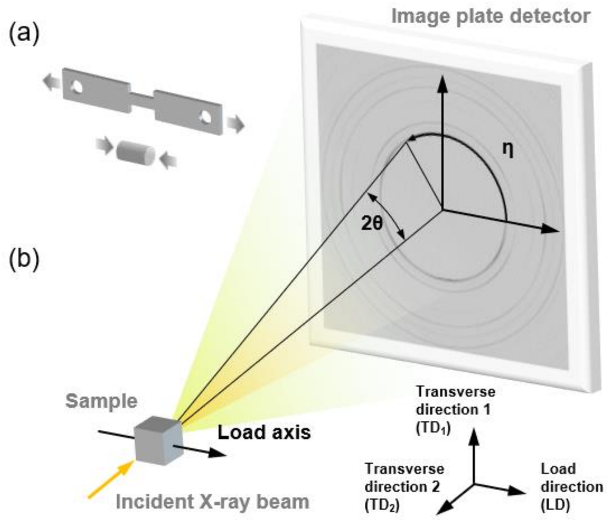


Figure 2: HEXRD setup. (a) Flat tension and round compression sample geometries. (b) The image plate detector behind the sample creates a cut in reciprocal space parallel to the load direction LD and sample transverse direction $TD_1$.

### 2.3. Crystal plasticity (CP) simulations

Texture evolution for uniaxial tension and compression was predicted by ALAMEL crystal plasticity simulations [46]. Simulations were performed for reference single-phase austenitic (fcc) and martensitic (body centred cubic – bcc) microstructures exhibiting a random crystallographic texture and 5000 spherical grains in the initial undeformed state. Slip on {111}<1-10> and on {110}<1-11> + {112}<11-1> systems were considered for austenite and for martensite, respectively, and all slip systems were assumed to be ideally plastic (non-hardening). Uniaxial tensile respectively compressive stress was imposed, the corresponding strain mode iteratively determined, and the deformation texture derived from the collection of evolving grain orientations [66]. Habit plane normals $\boldsymbol{n}$ and displacements $\boldsymbol{d}$ for the $\gamma \rightarrow \alpha$' MT were calculated from the phenomenological theory of martensite crystallography (PTMC) using PTCLab [67]. Crystal structure drawings were produced with VESTA 3 [68].

## 3. Macroscopic flow and hardening

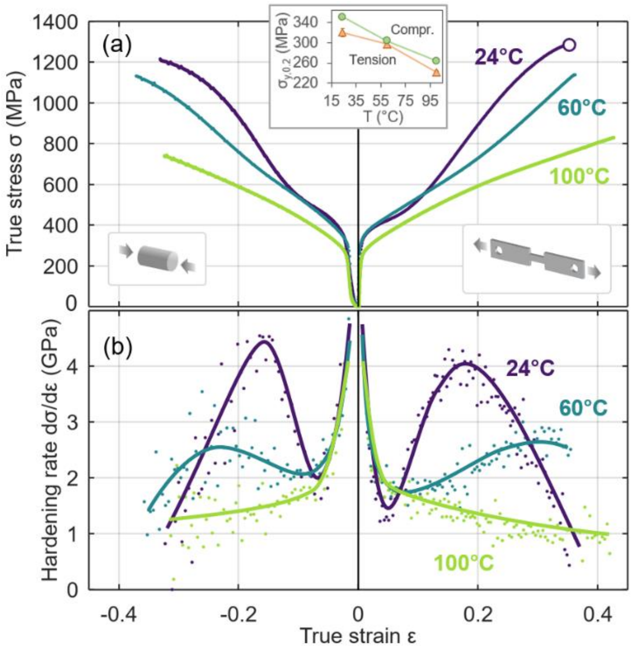


Figure 3: (a) Macroscopic stress-strain response of EN 1.4318 (AISI 301LN) polycrystals in uniaxial tension vs. compression at different test temperatures with the yield strength shown in the inset. (b) Hardening rates derived from curves in (a). Tension at 24°C was the only test that reached Considère's criterion for diffuse necking (indicated by the open circle).

Figure 3 shows macroscopic stress-strain curves and the attendant work hardening rates (WHR) for EN 1.4318 stainless steel tested in uniaxial tension and compression at 24°C, 60°C and 100°C. Hardening rates were calculated from true stress $\sigma$ and strain $\varepsilon$ as $\theta = d\sigma/d\varepsilon$. Increasing the test temperature from 24°C to 100°C reduces tensile and compressive yield strengths by 80-100 MPa. Not only are the general flow stress levels simultaneously reduced but also the hardening behaviour varies strongly with temperature. While the hardening rate decreases at low strain at all temperatures, it increases again after going through a minimum at 24°C and at 60°C. This behaviour leads to very high hardening rates > 4 GPa in compression and tension at 24°C for intermediate strains ($|\varepsilon| \approx 0.18$). At 60°C, the maximum WHR is considerably reduced relative to 24°C, namely to ~2.5 GPa. The maximum WHR is reached at comparably larger (absolute) strain (0.30 in tension and -0.23 in compression). At 100°C, the hardening rate decreases monotonically with deformation, which is reflected in the relatively low flow stress at this temperature (Figure 3a).

Several important features pertaining to the macroscopic deformation behaviour stand out:

(i) The test temperature strongly affects hardening and yield stress. Temperature dominates load sense effects (tension vs. compression) on the flow and hardening behaviour. The load sense exerts only a minor influence on the macroscopic mechanical behaviour.

(ii) Both stress-strain response and work hardening exhibit very weak TCA. However, in compression the maximum hardening rate at 24°C is slightly higher than in tension and it is reached at smaller strain. In the light of these observations, it deems more suitable to speak of (macroscopic) tension-compression symmetry instead of asymmetry.

(iii) The steel at hand exhibits a high resistance to macroscopic strain localization at all test temperatures. Only for tension at 24°C Considère's criterion for diffuse necking ($\theta = \sigma$) was reached at $\varepsilon = 0.33$. For the remaining tensile tests straining stopped at the maximum machine-controlled displacement before the onset of necking. The present results agree very well with room temperature tensile tests previously carried out on a similar steel of the same grade [6, 69].

As established in the next section, the temperature variation of plastic flow and work hardening correlates with the austenite instability towards deformation-induced martensite formation. A similar correlation is found between the variation of work hardening with plastic strain and the transformation rates.

## 4. Transformation behaviour

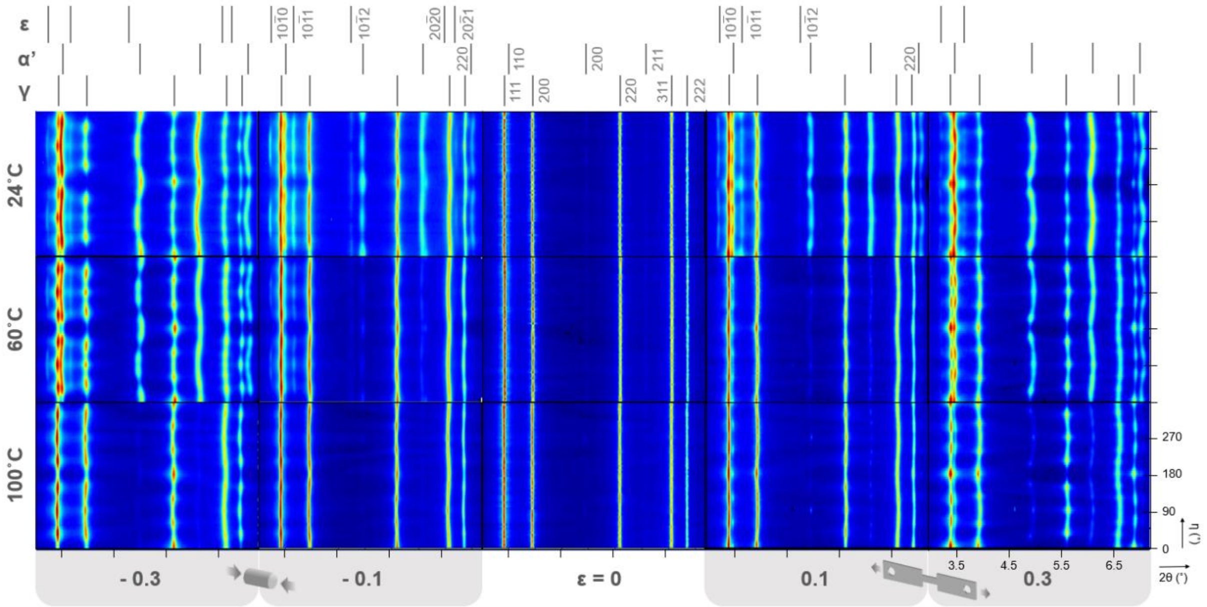


Figure 4: In-situ high-energy X-ray diffraction patterns in polar space of EN 1.4318 (AISI 301LN) polycrystals for uniaxial tension vs. compression at different temperatures and selected macroscopic true strains.

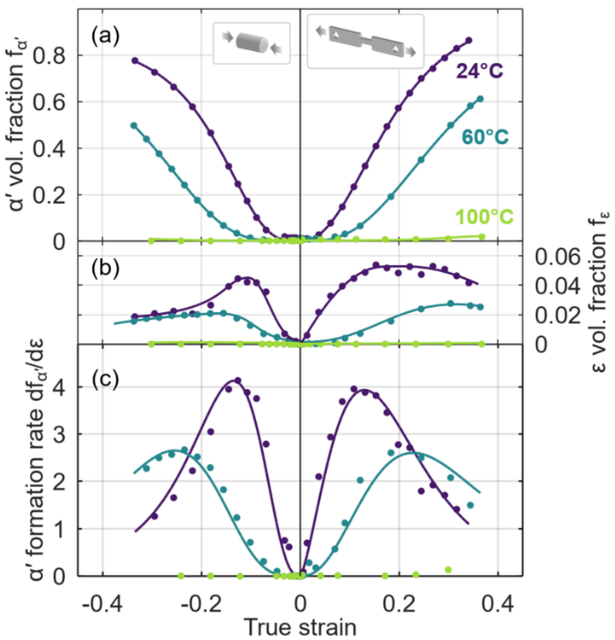


Figure 5: Evolution of the volume fractions of (a) α'-martensite and of (b) ε-martensite of EN 1.4318 (AISI 301LN) polycrystals during uniaxial tensile and compressive straining at different temperatures, and (c) the corresponding α' formation rate. Solid lines in (a) are Olson-Cohen fits discussed in Section 4.1.2.

Figure 4 compares in-situ recorded diffraction patterns for compressive and tensile absolute true macroscopic strains of 0%, 10% and 30% at different temperatures. These snapshots reveal a strong influence of temperature and load sense on (i) relative peak intensities of γ austenite and α' and ε martensites due to varying phase volume fractions, (ii) intensity variations along diffraction rings due to crystallographic texture, (iii) sinusoidal shifts of diffraction angles due to deviatoric type I internal strain. While this section focusses on the pertinent features of the transformation behaviour extracted from the global integrated peak intensities, the evolutions of texture and strain (stress) partitioning are demonstrated in sections 6 and 7.

#### 4.1.1. Martensite content

The evolution of the martensite fractions (α', ε) is presented in Figure 5. Independent of load sense, the transformation behaviour of both martensites is strongly impacted by temperature. At 24°C, the volume fraction of α' (denoted $f_{\alpha'}$) reaches almost 0.9 at a strain of 0.35. Increasing the temperature reduces that value to ~0.6 at 60°C and to <0.03 at 100°C. The underlying formation rate of α', presented in Figure 5c, depends strongly on temperature and strain. The highest transformation rates are found at 24°C, peaking at true strain $\varepsilon$ = 0.1-0.15. Heating to 60°C decelerates the transformation significantly and shifts the peak rates to larger deformations of $\varepsilon$ = 0.15-0.25. The evolution of $f_{\alpha'}$ exhibits only weak TCA, agreeing with the lack of asymmetry in the macroscopic flow and hardening behaviour.

Compared to α', much smaller amounts of ε martensite form, approximately 5% at 24°C and 3% at 60°C in tension (Figure 5b). In compression, ε formation becomes quickly exhausted and thus slightly less ε-martensite forms. No ε-martensite was detected at 100°C. Importantly, $f_\varepsilon$ goes through maxima for both load senses (for tension 60°C we anticipate the maximum at $\varepsilon$ = 0.35). In other words, while the amount of ε-martensite increases initially, it shrinks progressively with deformation. The start of the reduction of $f_\varepsilon$ is found at or close to the maximum α' formation rate for all temperatures and both load senses.

#### 4.1.2. α' kinetics

To further analyse the transformation kinetics we fitted Olson and Cohen's empirical model for $f_{\alpha'}$ as a function of plastic strain $\varepsilon_{pl}$ [70]:

$$f_{\alpha'}(\varepsilon_{pl}) = 1 - \exp\left\{-\beta\left[1 - \exp\left(-\alpha\varepsilon_{pl}\right)\right]^n\right\} \quad (1)$$

Equation 1 applies in quasi-static conditions and assumes that α' nucleation is governed by the occurrence of intersections of shear bands. Their nature is not further specified by the model, but in the present context is assumed to correspond to lamellas of ε martensite or stacking fault bundles. $\alpha$ represents the rate of shear banding, while $\beta$ is proportional to the probability that α' embryos nucleate in shear band intersections. Both $\alpha$ and $\beta$ are expected to be temperature dependent: $\alpha$ through the SFE (a smaller SFE at lower temperature leads to a larger α), and $\beta$ through the chemical energy difference of the martensitic phases. The exponent $n$ describes the formation rate of shear band intersections based on their geometric arrangement [45, 70] and has been often set to a constant value of 4.5 [20, 31, 41, 45, 70]. Unlike in those works, we found that $n =$ 4.5 did not give satisfactory agreement with the measured α' content, thus we included $n$ as a fit parameter. The best fitting parameters are compared in Figure 6 with values reported in the literature. The fitted curves are plotted together with the experimental data in Figure 5a.

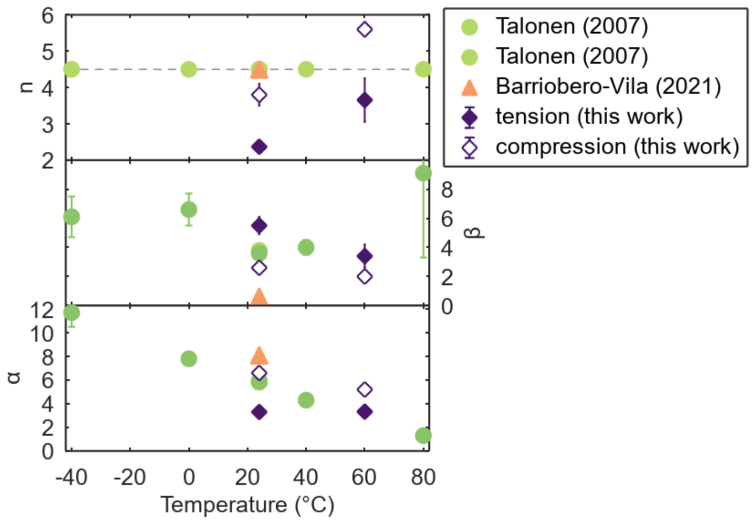


Figure 6: Parameters of the kinetic equation (Eq. 1) for strain-induced α’ martensite determined by least squares fitting. Literature values are for uniaxial tension at strain rates of (3−10)·$10^{-4}$/s.

Both $\alpha$ and $\beta$ tend to decrease with temperature, indicating less shear banding at increased temperature, in agreement with Talonen's data for $\alpha$ [45]. It further confirms early suggestions of β shrinking with temperature [20, 70] due to the reduced probability that ε shear band intersections form α’ martensite through the temperature dependence of the chemical driving force. Talonen's value for $\beta$ at 80°C seems to be an outlier, lying far off the decreasing trend and having a very large uncertainty [45]. In terms of TCA, our analysis shows that α is smaller in tension and β is smaller in compression. A similar observation was made by Iwamoto et al. for a 304 type MASS [31], suggesting that compression favours shear banding.

In contrast to the traditional assumption in the literature, the exponent $n$ increases with temperature, both in tension and compression. In tension, $n$ is much smaller than Olson and Cohen's value of 4.5 and in compression $n$ lies below and above 4.5. For randomly oriented, thin shear bands, quantitative stereology foresees $n$ = 2 [6]. Olson and Cohen [70] chose a constant $n$ = 4.5 by fitting the tensile test data of Angel [17] for an AISI 304 steel at different temperatures, arguing that shear bands tend to form parallel with each other, leading to $n \neq 2$. The present work demonstrates that $n$ = 4.5 does not hold for EN 1.4318 steel. Instead, Figure 6 suggests that shear bands at elevated temperature align stronger with each other than at room temperature. This situation corresponds to more co-planar deformation systems being active at elevated temperature. We now interpret the shear bands as ε martensite. For ε variants forming randomly inside a grain independent of temperature, $n$ is expected to be a constant value of 2. If however one ε variant dominates the other ε variants, $n$ > 2. Applying this interpretation to the present case suggests that compression as well as higher temperature favour the formation of parallel ε lamella, increasing $n$. Indeed, this conclusion is in line with the low ε martensite fraction at elevated temperature (Figure 5), assuming that fewer ε variants need to be active to produce low ε phase fractions. Plates of the first forming, dominant ε variant are aligned parallel with each other and only after nucleation of another, non-parallel variant intersections may form.

### 4.1.3. Transformation pathway TCA

Another intriguing aspect is the TCA of $f_\varepsilon$ (Figure 5b). In compression, $f_\varepsilon$ peaks quickly and decreases afterwards. In tension, $f_\varepsilon$ grows more slowly and peaks much later than in compression. This observation suggests that compression promotes the conversion of ε into α’ via the γ → ε → α’ pathway, where α’ martensite forms via ε martensite. The increased shear banding we found for compression in section 4.1.2 supports that the γ → ε → α’ route is stimulated by compressive loading. We ascribe this TCA to the interaction of the transformation volume change $|\boldsymbol{v}|$ with the applied load. As detailed in Appendix 9.1, the volume change for γ → α’ is +0.024 and for γ → ε it is -0.01 – -

0.0125 (Table 6). If we then posit that α’ always forms locally where it most favourably interacts with the applied load, the observed TCA in $f_\varepsilon$ then follows from volumetric strain energy minimization.

In tension, α’ most effectively interacts with the applied load when it forms directly from γ. Then the full volume dilatation can contribute to relaxing the external stress. However, when α’ forms instead from ε, part (namely about half) of the dilatation will only compensate the volume contraction adhering to ε, not contributing to relaxing the external stress. For compression the situation is different. Here, the overall internal strain energy can be reduced when α’ forms inside ε, since ε can compensate for some of the volume dilatation of α’. This finding is strengthened by recent micromechanical calculations of the transformation sequence in EN 1.4301 (AISI 304) by Wada et al. [49, 71]. Their results suggest that ε → α' provides a low (elastic) strain energy pathway for α' formation. Indeed, intersections of ε bands act as primary nucleation sites for α' at low strain rates [8, 10, 72]. α' has also been observed to form inside isolated ε lamellas [49, 73, 74], growing into and consuming ε-martensite on further staining. Consequently, the volume fraction of ε peaks rather quickly in compression, facilitating the formation and growth of α'-martensite.

## 5. Work hardening vs. transformation behaviour

Figure 7 compares the macroscopic WHR ($d\sigma/d\varepsilon$) with the transformation behaviour analysed by the 1st and 2nd derivates of the α' volume fraction. These correspond to the transformation rate $df_{\alpha'}/d\varepsilon$ and its change with strain, respectively. Only temperatures with significant amounts of α'-martensite are shown.

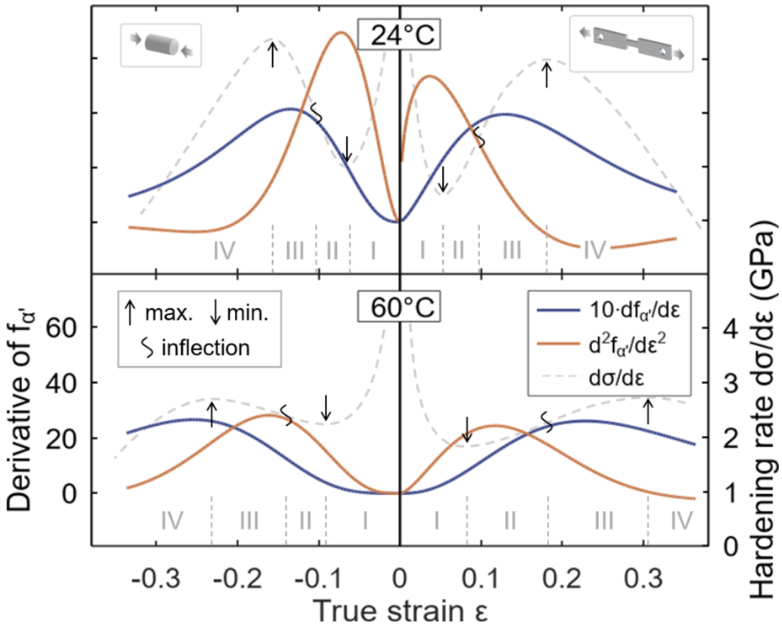


Figure 7: Correlation of the hardening rate with the 1st and 2nd derivatives of the α'-martensite content for EN 1.4318 (AISI 301LN) tested in tension and compression at 24°C and 60°C.

While it is widely reported that higher hardening correlates with a higher transformation rate, Figure 7 shows that this correlation holds only for maximum transformation and hardening rates, but not along the entire stress-strain curve, both in tension and in compression and at both temperatures. In tension at $\varepsilon$ = 0.04 and in compression at $\varepsilon$ = -0.07, the hardening rate is temporarily larger at high temperature despite a clearly lower transformation rate. Work hardening at low plastic strains is thus not only governed by the transformation rate, but other hardening mechanisms (such as slip interaction, twinning) must be considered to explain the hardening behaviour.

Based on the WHR, Talonen [45] divided the stress-strain evolution of MASS into four stages:

- Stage I: Initial section with rapidly decreasing WHR until the minimum WHR is reached. The decreasing WHR has been rationalized by dynamic softening due to nucleation of α'-

martensite in ε-martensite shear band intersections, also referred to as the "window effect". A very good correlation between maximum $d^2 f_{\alpha'}/d\varepsilon^2$ and minimum WHR was found [45].

- Stage II: The WHR is dominated by the generation of geometrically necessary dislocations in soft austenite γ, where hard α'-martensite islands are embedded, quickly growing in number and size. The inhomogeneous plastic deformation enhances work hardening of γ up to the percolation threshold of α'-martensite at $f_{\alpha'}$ = 0.3.
- Stage III: The beginning of stage III corresponds to formation of the first percolating cluster of α'-martensite and is linked to the inflection point of the WHR. Plastic deformation of the phase aggregate can no longer be accommodated by austenite alone, but martensite must co-deform. The WHR keeps growing, because of the high strength of martensite and its increasing volume fraction, among other factors.
- Stage IV: Characterized by decreasing WHR and reported to start at $f_{\alpha'}$ = 0.6–0.7. While the hardening capacity of austenite and martensite may not be fully exhausted, they cannot prevent strain localization and necking any longer.

While agreeing with Talonen's analysis in a broad sense, Figure 7 bears important differences in the correlation of the WHR with the transformation behaviour, where compressive deformation enlarges these deviations. These differences are discussed below.

First, the minimum WHR correlates well with the maximum $d^2 f_{\alpha'}/d\varepsilon^2$ only at 24°C. At 60°C, the minimum WHR precedes the maximum $d^2 f_{\alpha'}/d\varepsilon^2$ by $\varepsilon$ = 0.5–0.1. Thus, α' nucleation does not govern softening at elevated temperature. More likely responsible is a combination of factors including lowered austenite critical resolved shear stress (CRSS) and enhanced dislocation mobilities. Second, the inflection point of the WHR (separating stage II from stage III) always comes significantly before the maximum of the transformation rate $df_{\alpha'}/d\varepsilon$. Transitions between stage II and III in Figure 7 occur at $f_{\alpha'}$ = 0.15–0.2 for 24°C and at $f_{\alpha'}$ = 0.05–0.2 for 60°C, i.e. below the α' martensite percolation threshold $f_{\alpha'}$ = 0.3 of postulated by Talonen to control the WHR inflection [45]. Especially in compression the inflection point of the WHR lies far below $f_{\alpha'}$ = 0.3.

The present data (Figure 5 and Figure 7) suggest that the rising flow stress in stage II is primarily controlled by work-hardening and dispersion-strengthening of austenite. The α' volume fraction has only an indirect role. Indeed, the flow stress of a two-phase composite consisting of a soft matrix and hard dispersions is mainly governed by the flow stress of the matrix phase [75]. Stage III starts when the increase of the WHR does not further speed up. Assigning this feature to a single criterion like the martensite percolation threshold is problematic, given the multitude of hardening mechanisms occurring simultaneously. Our observation that the α' volume fraction only partially influences the hardening stages is corroborated by the transition from stage III to stage IV. It takes place at $f_{\alpha'}$ = 0.4–0.5 at 24°C and at $f_{\alpha'}$ = 0.25–0.5 at 60°C, in other words clearly below the $f_{\alpha'}$ = 0.6–0.7 reported by Talonen [45]. Comparable to the inflection of the WHR, the WHR maximum is controlled by the microstructure's ability to harden and to prevent strain localization. This ability is affected by temperature and therefore stage IV commences at a lower $f_{\alpha'}$ at elevated temperature.

## 6. Texture evolution

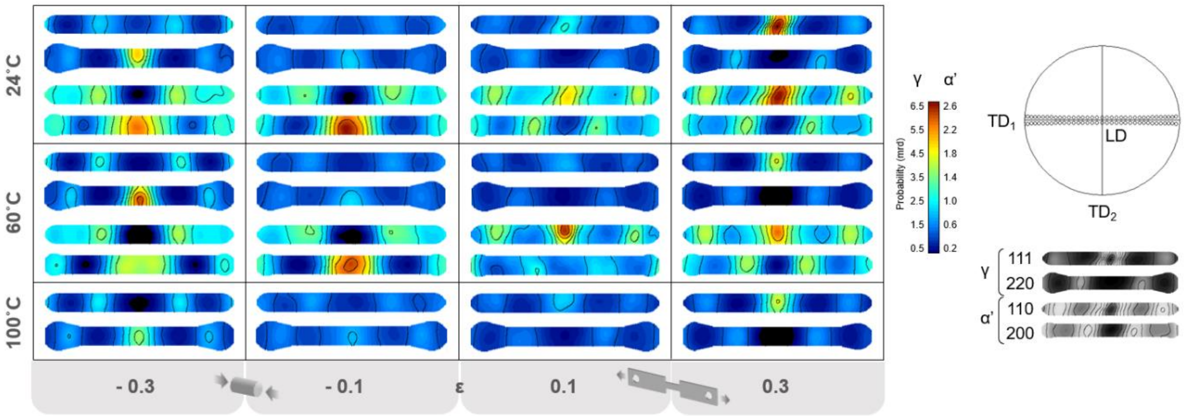


Figure 8: Temperature dependent texture evolution of austenite (γ) and α'-martensite in tension vs. compression measured by in-situ HEXRD. For 24°C and 60 °C top and bottom pole figures are for γ and α', respectively. For 100°C only γ pole figures are shown because of the negligible volume fraction of α'. Pole figures cover orientations in close vicinity of the LD-$TD_1$ plane.

Figure 8 shows snapshots of the texture evolution in terms of measured pole figures for the three test temperatures in tension vs. compression. The pole figures derived from the in-situ recorded 2D HEXRD patterns cover orientations close to the plane spanned by loading and one transverse direction (LD-$TD_1$), corresponding to the pole figure equator. The centre of the pole figures aligns with the load axis LD and thus illustrates the change of the fibre components formed along LD. Because of the uniaxial character of the imposed load and the near randomness of the starting texture, we expect axisymmetric textures to form, rotationally symmetric about LD. Figure 8 confirms our expectation and the following analysis therefore concentrates on texture fibres along LD.

Figure 8 reveals a strong TCA in texture development, both for austenite and α'-martensite, largely unaffected by temperature: all temperatures evidence similar trends apart from slight changes in fibre intensities. In tension, the $<111>_\gamma$ and $<110>_{\alpha'}$ fibres grow with strain while $<110>_\gamma$ and $<100>_{\alpha'}$ fibres weaken[1]. Compression largely inverts the texture evolution observed in tension. $<111>_\gamma$ and $<110>_{\alpha'}$ fibres weaken while $<110>_\gamma$ and $<100>_{\alpha'}$ fibres strengthen with compressive strain.

#### 6.1.1. In-situ observations vs. crystal plasticity simulations

Figure 9 quantifies the measured fibre strength depending on strain, load sense and temperature and compares them with CP simulations. These simulations were carried out separately for austenite and for α'-martensite assuming plasticity to occur by conventional dislocation slip without MT. The simulated fibre strengths (Figure 9b) thus provide theoretical baselines to identify texture features due to the simultaneous occurrence of grain rotation due to dislocation slip *and* of MT in transforming samples.

Both plastic slip and MT determine texture evolution as demonstrated by comparison of measurements with simulations in Figure 9. For tension, CP simulations predict a strong $<111>_\gamma$ fibre next to a weak $<100>_\gamma$ fibre, and a gradually vanishing $<110>_\gamma$ fibre. Such double fibre $<111>_\gamma + <100>_\gamma$ textures are typical for fcc materials deformed by axisymmetric elongation, including uniaxial tension, wire drawing, extrusion and round swaging [76, 77]. For compression, simulations predict fibre intensities reversed relative to tension: a weak $<110>_\gamma$ fibre and vanishing $<111>_\gamma$ and $<100>_\gamma$ fibres, as expected for uniaxially compressed fcc polycrystals [76, 77].

[1] $<110>_\gamma$ and $<100>_{\alpha'}$ fibres were determined from $\{220\}_\gamma$ and $\{200\}_{\alpha'}$ reflections, respectively.

Textures measured at 100°C agree with the simulations, confirming the absence of MTs at that temperature. On the other hand, textures at 24°C and 60°C bear clear signatures of the strain-induced MT in comparison with transformation free textures:

- In tension, the $<100>_\gamma$ fibre intensifies in absence of the MT, but it weakens as soon as the MT occurs. Consequently, at 60°C the $<100>_\gamma$ fibre first intensifies at low strain where no to little martensite forms, but it wanes as soon as $f_{\alpha'} > 0.1$. At 24°C the fibre loses strength already from low strains on, due to the high propensity to form martensite. A weakening of $<100>_\gamma$ orientations in the presence of γ → α' MT was also reported in tension tests on Cr-Ni-Mo MASS [28] and TRIP-assisted steel [42, 78], resulting in austenite textures dominated by $<111>_\gamma$ grains.
- In compression, the $<111>_\gamma$ and $<100>_\gamma$ fibres decrease for deformation by slip, where $<111>_\gamma$ tends to weaken more than $<100>_\gamma$. In the presence of the MT, the $<111>_\gamma$ fibre weakens less (at 60°C) and even slightly increases (24°C).

#### 6.1.2. Linking texture TCA and γ → α' driving force anisotropy

To investigate the origin of these texture features, we study the mechanical driving force *U* for the γ → α' MT depending on orientation and direction of the load axis (see Appendix 9.2). Figure 9c shows the resulting driving force *U* per MPa imposed stress for the three direction families. For positive and negative values of the driving force, uniaxial tension and compression activates the habit plane variant (HPV), respectively. The magnitude of the driving force indicates the ease with which the HPV forms, i.e., large $|U|$ favours the MT and grain orientations with small $|U|$ will transform last.

Independent of load direction, $<100>_\gamma$ oriented grains experience the largest driving force, followed by $<110>_\gamma$ grains (Figure 9c). Orienting the stress axis along $<111>_\gamma$ results in the smallest driving force. These results compare well with calculations by Hilkhuijsen et al. [28], who analysed the driving force for Cr-Ni-Mo MASS under tension. Figure 9c shows that grains with a $<100>_\gamma$ direction along the load axis need about four times less stress to reach the critical driving force for γ → α' compared to $<111>_\gamma$ orientated grains, both in tension and in compression. Consequently, a $<100>_\gamma$ fibre will be weakened four times stronger by the MT than a $<111>_\gamma$ fibre. A $<110>_\gamma$ fibre will be reduced almost two times and four times as fast as a $<111>_\gamma$ fibre in tension and compression, respectively. Furthermore, the part of *U* linked to the shear ***s*** along the habit plane dominates the driving force; the dilatational part ***v*** due to the increased atomic volume in α' is of little significance.

The texture evolution during uniaxial loading (Figure 9a) reflects this anisotropy in the driving force of the γ → α' MT (Figure 9c). The high transformation propensity of $<100>_\gamma$ oriented grains weakens the $<100>_\gamma$ fibre relative to $<110>_\gamma$ and $<111>_\gamma$ fibres, both in tension and in compression. Another consequence of the driving force anisotropy is that the relative strengths of $<111>_\gamma$ and $<100>_\gamma$ compression fibres (Figure 9a) are inverted compared to simulations (Figure 9b). These effects are overlaid on the slip-only texture evolution predicted by CP simulations.

Comparison of in-situ results and CP simulations reveals that not plastic deformation but MT and the attendant driving force anisotropy control α' martensite texture evolution. CP simulations predict a continuous increase of $<110>_{\alpha'}$ fibre in tension and of $<100>_{\alpha'}$ fibre strengths in compression. In contrast, measurements show a weakening of these fibres after an initial strengthening. The evolution of α' martensite texture is largely controlled by the MT itself, i.e., the orientation relationship coupled to the transformation propensity make the observed martensite texture emerge. To verify this proposition, we use the PTMC to predict the dominant α' texture fibres considering the anisotropy of the MT driving force, drawing parallels with the in-situ observations.

For each of the three austenite fibres ($<100>_\gamma$, $<110>_\gamma$, $<111>_{\gamma,}$), Table 4 lists the largest driving forces for tensile and compressive loading and transforms the pertaining load directions from austenite to α' martensite for a representative HPV (***d*** = [0.156, -0.159, 0.046], ***n*** = (-0.608, -0.774,

-0.178)). The nature the LIS of α’ martensite in EN 1.4318 is mostly slip [6, 23]. The HPV thus largely consists of one martensite variant. Each load direction in austenite then corresponds to one direction in martensite, calculated from the PTMC and given in the last column of Table 4. These directions are the transformation fibre textures to form from the parent austenite fibre textures by uniaxial stress loading.

The $<100>_\gamma$ fibre gives rise to the $<110>_{\alpha'}$ fibre in tension and to the $<100>_{\alpha'}$ fibre in compression. $<100>_\gamma$ experiences the largest driving forces out of the three austenite fibres and hence transforms most easily both in tension and in compression. Indeed, both related martensite fibres came out strongest (Figure 9a). The austenite fibre experiencing the second largest driving forces, $<110>_\gamma$, inherits the second strongest martensite fibres for both load senses ($<100>_{\alpha'}$ in tension and $<211>_{\alpha'}$ in compression). Eventually, the $<111>_\gamma$ fibre, experiencing the weakest driving forces, gives rise to some of the weakest martensite fibres ($<210>_{\alpha'}$ in tension and $<110>_{\alpha'}$ in compression).

The established links between austenite and martensite textures then reveal the origin for the clear weakening of the $<100>_{\alpha'}$ fibre relative to other martensite fibres in compression. Its parent fibre, $<100>_\gamma$, is not only the austenite fibre most favoured for MT but it also weakens by slip (Figure 9b). Hence, while initially propelling the MT, the $<100>_\gamma$ fibre is quickly exhausted. On the other hand, grain rotation due to slip continuously supplies new austenite grains for the $<110>_\gamma$ fibre and its transformation fibre, $<211>_{\alpha'}$, grows relative to $<100>_{\alpha'}$.

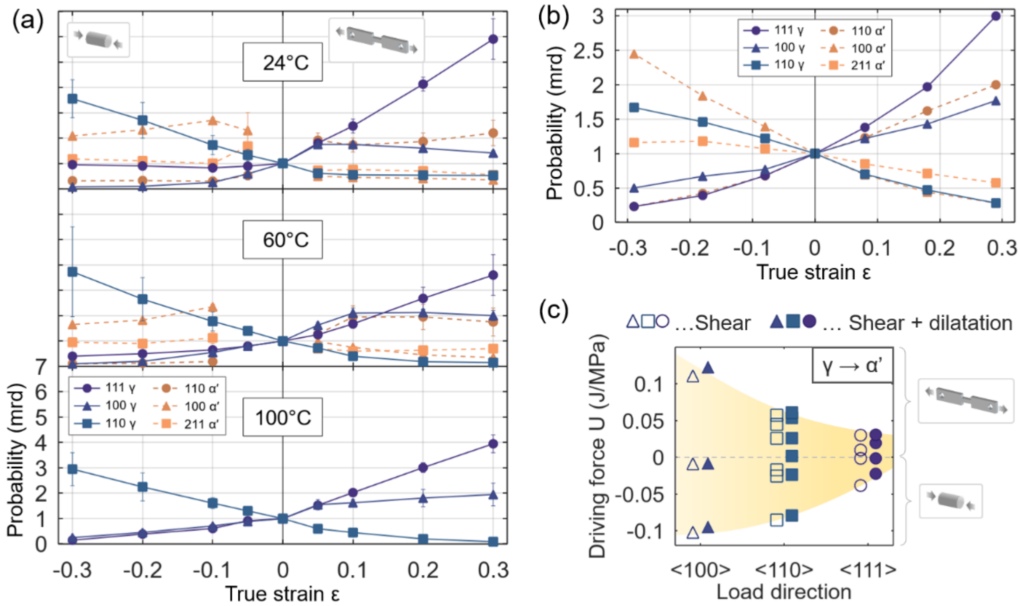


Figure 9: (a) Temperature dependent evolution of texture fibres along the load axis for γ austenite and for α'-martensite in tension vs. compression measured by in-situ HEXRD. (b) Evolution of the same fibres predicted by CP simulations for single-phase γ and α’ deforming by slip only. (c) Directional dependence of the driving force for γ → α’.

Table 4: Transformation of austenite texture fibres exhibiting largest driving forces in tension and compression into α’ martensite fibres.

| Austenite fibre | Load sense | U | Load axis Austenite | α’ martensite | |
|---|---|---|---|---|---|
| | | mJ/MPa | | exact | approx. |
| <100> | Tension | 123 | [0,1,0] | [-0.63, 0.77, 0.05] | [-1,1,0] |
| | Compression | -95 | [0,0,1] | [-0.13, -0.17, 0.98] | [0,0,1] |
| <110> | Tension | 61 | [1,1,0] | [0.09, 0.98, 0.18] | [0,1,0] |
| | Compression | -79 | [1,0,1] | [0.45, 0.31, 0.84] | [1,1,2] |

| | | | | | |
|---|---|---|---|---|---|
| <111> | Tension | 31 | [-1,1,1] | [-0.88, -0.01, 0.48] | [-2,0,1] |
| | Compression | -22 | [1,-1,1] | [0.73, -0.19, 0.65] | [1,0,1] |

## 7. Strain/Stress sharing between austenite and martensites

The undeformed polycrystals ($\varepsilon$ = 0) are free of deviatoric type I residual strain, as the $\eta$-independent diffraction angles evidence (Figure 4). The corresponding unit cell parameters of γ austenite and δ ferrite increase slightly with temperature due to thermal expansion (Table 5). Calculation of the lattice strain faces the intrinsic difficulty that strain-induced martensite cannot be obtained in a completely stress-free state [69], yet a certain choice for reference state has to be made. Here, we use as reference the bcc unit cell at ~5 vol.% strain-induced α', next to the austenite parameters in Table 5.

Table 5: Unit cell parameters of austenite and of δ ferrite prior to deformation at the three test temperatures.

| Temperature (°C) | $a_\gamma$ (Å) | $a_\delta$ (Å) |
|---|---|---|
| 24°C | 3.58932(3) | 2.870(2) |
| 60°C | 3.59088(3) | 2.871(1) |
| 100°C | 3.59306(3) | 2.873(3) |

Figure 10a,b shows the lattice strains measured via in-situ HEXRD for austenite and α' martensite depending on macroscopic uniaxial true stress and on tensile true strain. From the initial load segments (Figure 10b) we can see that γ plastifies before α' formation sets in. To extract critical insight into TRIP-specific micromechanics due to dynamic strain (stress) sharing between austenite and martensites, the global behaviour of the lattice strains caused by the intrinsic anisotropy of Young's modulus must be understood first.

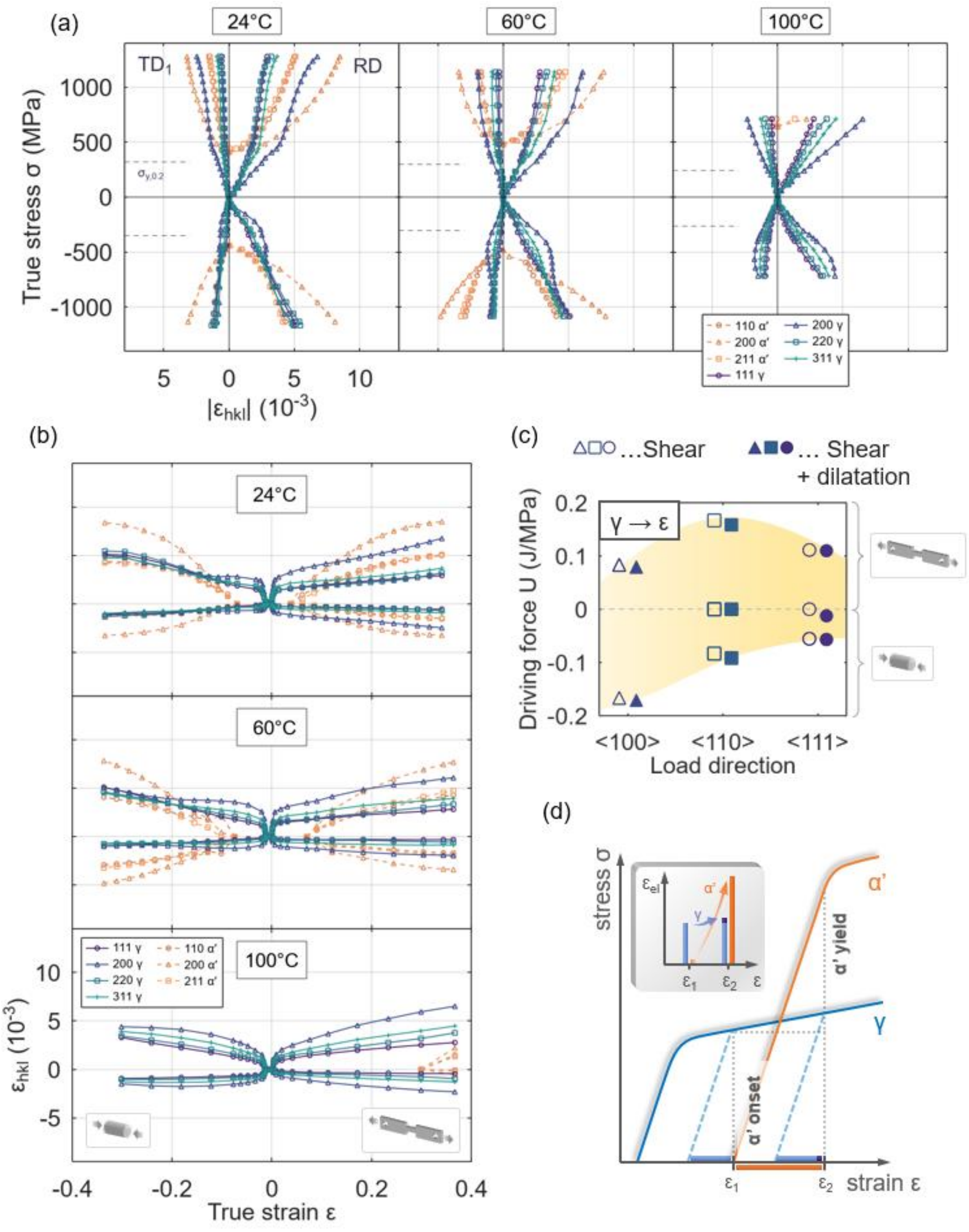


Figure 10: Tension-compression asymmetry of lattice plane-specific strain partitioning between γ austenite and α'-martensite at 24°C, 60°C and 100°. (a) Influence of macroscopic true stress σ. (b) Influence of macroscopic true strain ε. (c) The driving force for γ → ε depending on load direction and sense. (d) Stress sharing between γ and strain-induced α'.

#### 7.1.1. Influence of $E_{hkl}$

Comparison of the plane specific strains and the directional elastic moduli of fcc γ [79] and bcc α' [80, 81] reveals that Young's modulus $E$ governs the relative order of the measured lattice strains at all temperatures and both load senses. For both γ and α’ $E_{100} < E_{113} < E_{112} \approx E_{110} \approx E_{111}$ holds. The lattice strains exhibit the inverse order $|\varepsilon_{100}| > |\varepsilon_{113}| > |\varepsilon_{112}| \approx |\varepsilon_{110}| \approx |\varepsilon_{111}|$. Thus, the lower $E_{hkl}$ the larger $|\varepsilon_{hkl}|$, and vice versa. Mostly the same order of $|\varepsilon_{hkl}|$ was found for tension at room temperature for non-transforming fcc stainless steel [82, 83], AISI 201 MASS [84], predominantly ferritic TRIP steel [85], as well as for tension below RT for low-alloyed TRIP-assisted steel [42]. Our in-situ measurements reveal that the inverse relationship between $E_{hkl}$ and $|\varepsilon_{hkl}|$ also holds for compression (Figure 10). However, within this general behaviour dictated by $E_{hkl}$, lattice strains vary prominently with temperature and load sense. In absence of MTs, as is the case for tension at 100°C, rather linear relationships are expected between γ lattice strains and applied stress, with the plane-specific Young’s modulus $E_{hkl}$ as the proportionality factor. In the presence of MTs, γ strains become redistributed among grain families and transferred to martensite crystallites, leading to more complex variations of the lattice strains with applied stress. They reveal critical insight into the micromechanics of MASS transforming under loading and document the important role of directionality and volume change of the MTs in MASS.

#### 7.1.2. Lattice strain TCA due to γ → ε and γ → α’

γ lattice strains at 24°C and 60°C exhibit prominent TCA:

- Under tension, γ strains fan out in the inverse order of $E_{hkl}$ and their spread increases with applied stress/strain across the entire stress-strain curve. The appearance of α’ slows down further strain accumulation on all γ grain families (Figure 10a).
- Under compression, γ strains initially also fan out as in tension, their spread increasing with applied stress (strain). But *before* widespread α’ formation sets in, strain accumulation slows down only on soft planes (having small $E_{hkl}$) but not on stiff planes (having large $E_{hkl}$). Notably, strain accumulation on $\{111\}_\gamma$ temporarily ceases. γ lattice strains eventually become equal and stay almost equal for the rest of the stress-strain curve while continuing to increase.

Note that under tension, retardation of γ strain accumulation coincides with the start of α’ formation, while under compression retardation begins prior α’ formation. In both cases, γ has already yielded plastically, see Figure 10b, hence strain retardation in the $\sigma$ vs. $|\varepsilon_{hkl}|$ plots in Figure 10a is not linked to austenite plastification, but must be of different origin such as the MTs.

On the continuum level, a MT manifests as shear parallel to the habit plane and dilatation perpendicular to the habit plane. From our in-situ HEXRD measurements, we calculated shear and dilatation for γ → α’ and for γ → ε (see Appendix 9.1). For γ → α’ we obtained a volume increase of 2.4%, and a volume contraction of 1 – 1.25% for γ → ε. While the volume expansion upon α’ formation is well documented in the literature [6, 24, 86], recent studies confirm the here observed volume shrinkage related to ε formation [49, 71].

The volume changes of γ → α’ and of γ → ε would thus relax a tensile and a compressive load, respectively. To get a full picture, the mechanical driving force $U$ for the MTs must also be taken into account because it controls how likely different γ grain families transform (Figure 9c and Figure 10b, see Appendix 9.2 for its calculation). While for each family of load direction, some combinations of HPV and load direction lead to very small $U$ close to zero, the combinations leading to the largest (for tension) and the smallest (i.e. most negative, for compression) must be examined here. These correspond to the positive and negative envelopes of the driving force plots.

Comparison of Figure 9c and Figure 10b shows that in compression, $|U_{\gamma\rightarrow\alpha'}| < |U_{\gamma\rightarrow\varepsilon}|$ for all orientations and $|U_{\gamma\rightarrow\varepsilon}|$ is largest for $\langle 100\rangle_\gamma$ followed by $\langle 110\rangle_\gamma$ and $\langle 111\rangle_\gamma$. Notably, the extent of the observed strain retardation in compression follows the same order, i.e. it is largest for $\{100\}_\gamma$ followed by $\{110\}_\gamma$ and by $\{111\}_\gamma$. This correlation strongly suggests that the driving force anisotropy and the volume change of the γ → ε MT are responsible for the unusual lattice strain evolution in compression. Indeed, formation of ε martensite coincides with the (macro) true strain range of the retardation phenomenon, as the plots of the volume fraction $f_\varepsilon$ in Figure 5b reveal. In tension, $U_{\gamma\rightarrow\alpha'} > U_{\gamma\rightarrow\varepsilon}$ only for $\langle 100\rangle_\gamma$ oriented grains. For other grain families $U_{\gamma\rightarrow\alpha'} < U_{\gamma\rightarrow\varepsilon}$. In fact, for $\langle 110\rangle_\gamma$ and $\langle 111\rangle_\gamma$ grains $U_{\gamma\rightarrow\varepsilon}$ is 3 – 4 times larger than $U_{\gamma\rightarrow\alpha'}$, favouring γ → ε over γ → α’. Significant amounts of ε martensite hence form in tension (Figure 5b). However, the volume contraction linked to ε formation works against the applied tensile load, creating tensile stresses in the surrounding γ matrix and preventing γ lattice strains to relax in the same manner as in compression.

A weak TCA in the γ lattice strains is also observed at 100°C. Macroscopic stress and lattice strains follow linear relationships for tension (Figure 10a), evidencing γ plasticity and resultant strain hardening without phase transformations [87]. However, under compression strain accumulation on $\{111\}_\gamma$ slightly retards from $\varepsilon \approx 0.15$ onwards, which reminds of $\{111\}_\gamma$ relaxation due to γ → ε at 24°C and 60°C. This suggests that some ε martensite also formed at 100°C in compression even though it could not be reliably picked up by HEXRD.

#### 7.1.3. Strain sharing and the influence of initial martensite content

The in-situ lattice strain measurements present a different strain evolution for α' compared to γ (Figure 10). In contrast to α', sudden kinks as those due to plastic yielding and selective strain relaxation due to ε formation are absent. Instead, α' lattice strains exhibit evenly decreasing slopes both in $\sigma$-$|\varepsilon_{hkl}|$ plots and in $|\varepsilon_{hkl}|$-$\varepsilon$ plots. α' strains grow rapidly from the moment the first α' has formed and compliant martensite planes such as $\{200\}_{\alpha'}$ become easily strained more than the most compliant $\{200\}_\gamma$. This behaviour is consistent with published data for other MASS [69, 87], but unlike ferritic(-bainitic) TRIP-assisted steel, where the bcc lattice strain did not reach more than 50% of the fcc strains [78, 85]. We ascribe this to the relatively small fractions (<10%) of α' formed in TRIP-assisted steel, leaving large amounts of matrix material consisting of ferrite and retained austenite that can plastically flow around the martensite volumes. Secondly, the TRIP-assisted steels in those works exhibit low uniform elongations, precluding uniaxial deformations comparable to those of MASS where strain-induced α' can develop larger strains.

Note, that the α' lattice strains in Figure 10a,b behave differently from when α' is present prior to loading, like in the works by Spencer et al. [88, 89], who loaded γ-α' two-phase microstructures containing  up to 67% α'. In that case, austenite and martensite deform together purely elastically until the softer austenite yields by dislocation slip. The ensuing progressive yielding leads to an extended elastoplastic transition and high apparent hardening, which has been described by Masing-type models [88]. The present situation on the other hand, where loading starts from a fully austenitic microstructure and where α' forms only once γ has plastified, may be schematized as in Figure 10d. Until the start of α' formation ($\varepsilon_1$ in Figure 10d), elastic and plastic deformation is fully carried by γ. Freshly formed α' deforms elastically until its yield stress is reached. Having considerably higher strength than γ [40] but relatively similar $E$ [69, 88], α' can take up significantly larger elastic strains than γ. That combined with the limited work hardening of γ, α' strains rapidly overtake γ strains. Once α' has yielded ($\varepsilon_2$ in Figure 10d), γ and α' co-deform plastically.

After the onset of α' formation, the imposed strain can be simultaneously accommodated by at least three anelastic mechanisms, giving rise to the observed continuous retardation of accumulation of α' lattice strain: (i) γ plasticity by dislocation generation, multiplication and slip, (ii) shear and volumetric strains linked to the MT itself (Table 6), (iii) plastic deformation of α' [69, 88]. We propose the first two mechanisms (γ plasticity, transformation strains) to dominate, because they clearly occur as evidenced by the γ lattice strains (Figure 10b) and the increasing $f_{\alpha'}$ (Figure 5a). Judging from a comparison of the hardening curves (Figure 3) with those in the work of Spencer et al. [88] for a similar MASS, we conclude α' plasticity to be of little importance.

#### 7.1.4. Stress sharing and the effect of the γ → α' volume expansion

To analyse stress partitioning between austenite and α' martensite the diffraction-based lattice strains must be converted into stresses. Per phase, we calculated the average lattice parameter $\langle a \rangle$ across all reflections along LD and the corresponding phase strain from the variation of $\langle a \rangle$ with imposed load. We then obtained phase-specific stresses by multiplying the phase strains with the Eshelby–Kröner estimates of polycrystal Young's moduli for γ ($E$ = 196 GPa) and α' ($E$ = 213 GPa) [69]. Fits of the in-situ measured phase strain vs. $\sigma$ curves gave a Young's modulus for γ consistent with [69, 88] at both temperatures. Note that due to the unequal stress sharing between γ and α', similar fits for α' at the onset of α’ formation are biased leading to an unrealistic modulus and hence we used the values mentioned above derived from single crystal data.

The specific stress carried by γ and by α’ martensite is shown in Figure 11a. Note that these curves are independent of phase fractions. While the stress evolution for α’ is largely independent of load sense, it reveals a substantial TCA for γ at the temperatures where α’ formed. At low true strain, in the absence of α’, γ stress is symmetric. From this we infer that the dislocation arrangements in γ due to plastic flow are equivalent in tension and in compression, leading to the same hardening behaviour for both load senses. The TCS of the macroscopic flow curves at 100°C strengthens this idea (Figure 3a and Figure 11a). However, shortly after the onset of α’ formation the flow stress in γ increases more rapidly in compression than in tension. Because the hardening behaviour for γ is the same in tension and in compression (as evidenced by the flow stress symmetry at low strain), this means that at the same macroscopic true strain $|\varepsilon|$, γ must have experienced more plastic deformation in compression compared to in tension. The origin can be likely found in the local elasto-plastic accommodation of the volume expansion linked to the phase transformation [90], called Greenwood-Johnson effect [91]. In compression, the volume expansion during α’ formation works against the external load, locally raising the internal stress at transformation front. Thereby, stress in the relatively weak austenite can increase above the critical resolved shear stress, stimulating the generation of dislocations in austenite layers surrounding the growing α’ volumes [40, 92]. In tension on the other hand, the volume expansion relaxes the internal stress in γ.

Once α’ martensite has formed, it rapidly takes up load and quickly carries a higher stress than austenite. This result is consistent with the lattice strain evolution in Figure 10 and the common idea that α’ volumes act as stiff and strong elastoplastic reinforcements, similar to second phase particles in dispersion strengthened materials [89, 93]. In the present case, α' stress becomes larger than γ stress for $f_{\alpha'} \approx 0.25$ in compression and for $f_{\alpha'} \approx 0.15–0.2$ in tension, i.e., the cross-over of the phase-specific stresses is delayed for compression, in part due to the observed accelerated hardening of austenite during upsetting.

Figure 11b shows the total stress carried by each phase, i.e., the specific stress from Figure 11a weighted with the phase fraction. The stress contribution of α’ decreases with temperature while that of γ goes up because of the lower amount of α’ forming at high temperature. Once the MTs have commenced, the total stress borne by γ starts falling quickly, despite the prominent strain-hardening of γ at both temperatures and load senses. Comparison of the two load senses shows that austenite contributes a larger share to the total stress in compression than in tension. This asymmetry can be explained by the stronger elasto-plastic accommodation of the volume expansion due to α’ formation in compression, as discussed in Figure 10a. α’ still carries 75% and 50% of the total stress in compression, compared to 90% and 60% in tension at 24°C and 60°C, respectively, at a true strain $|\varepsilon| = 0.3$.

Another important conclusion that can be drawn from Figure 11b is that ε-martensite can contribute significantly to the macroscopic stress, despite its small volume fraction reaching not more than 5%. We estimated the stress carried by ε as the difference between the total true stress on the sample and the sum of the phase stresses for γ and α’. ε contributes up to 120 MPa and up to 20% of the total stress depending on test condition and applied strain. Thus, ε not only facilitates α’ formation (see section 4.1.3) but also enhances the macroscopic hardening rate further.

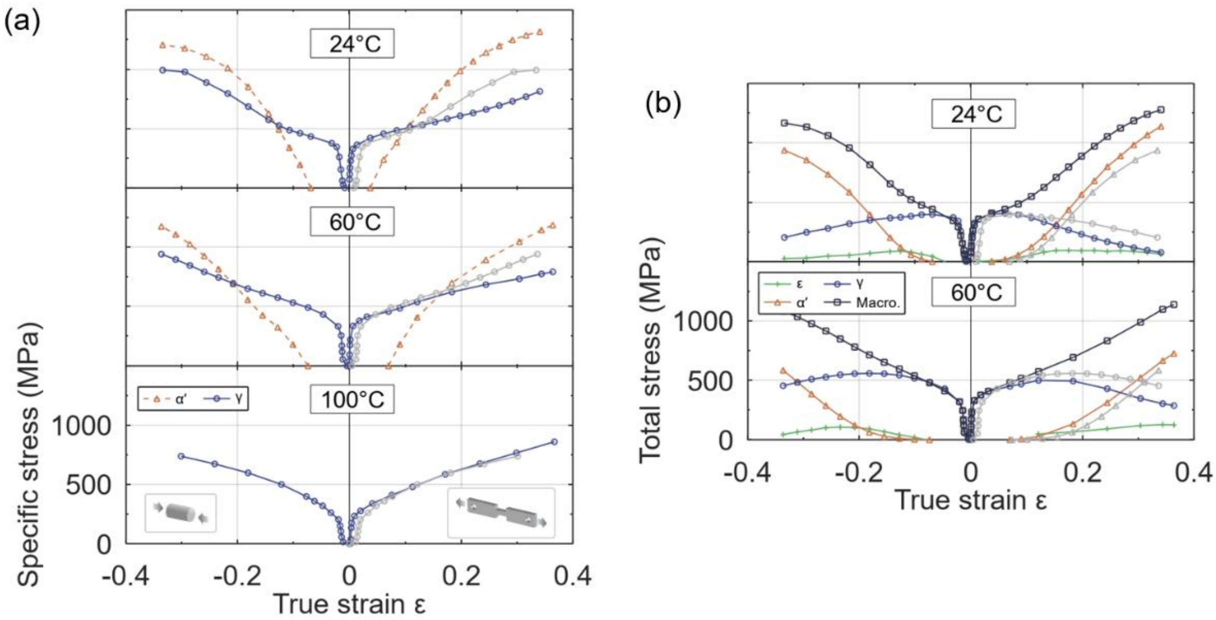


Figure 11: Stress partitioning for tensile and compressive loading of EN 1.4318 stainless steel depending on temperature. (a) Phase specific stress for austenite and α'-martensite. (b) Sharing of the macroscopic stress between austenite (γ) and martensites (α', ε). The compressive behaviour of γ (a) and of γ and α' (b) is mirrored onto the tensile side in grey.

## 8. Summary and Conclusions

The sense of uniaxial loading has only a minor influence on the macroscopic mechanical behaviour of EN 1.4318 MASS. By contrast, the test temperature strongly affects yield stress and work hardening, dominating load sense effects. At 24°C and 60°C, both α' and ε martensites form, while 100°C creates only a negligible amount of α'. Due to the MTs at 24°C and 60°C the WHR peaks at intermediate true strain, while at 100°C the WHR continuously decreases with deformation.

The lack of macroscopic stress-strain TCA can be explained by the closely matching evolution of α' martensite content in tension vs. compression. However, while the volume fraction of α' martensite increases continuously, the γ → ε → α' transformation pathway found for both load senses makes the amount of ε-martensite grow only initially before shrinking with progressing deformation. Compression promotes the conversion of ε into α', suggesting that γ → ε → α' provides a low (volumetric) strain energy pathway for α' formation.

While the notion that more martensite forming leads to higher hardening holds up in general terms, purely mathematical features (such as minimum, maximum, inflection point) of transformation kinetics and of work-hardening curves can neither be conclusively linked to each other nor to the stress-strain response. Clear correlations between these features seem to be a result of coincidence, limited to a specific steel composition, test temperature and tensile deformation. Instead, the temperature dependence of the austenite CRSS and SFE, the influence of composition and strain mode on transformation pathway and kinetics need to be considered.

The strong TCA in the evolution of crystallographic texture bears signatures of plastic slip and MT, and their dependence on load sense. Texture evolution of austenite is governed both by grain rotation due to dislocation slip and by MT. However, the MT loses relevance as austenite stability increases with temperature, as the CP simulations show. On the other hand, texture evolution of α' is largely controlled by the MT itself, not by plastic deformation. The orientation relationship between γ and α' coupled to the directionality of the transformation make the observed martensite texture emerge. Combining the PTMC and the driving force anisotropy for α' formation is an effective approach to predict the relative strengths of austenite and martensite texture fibres, for both load senses.

The inverse relationship between $E_{hkl}$ and $|\varepsilon_{hkl}|$ found through the analysis of differently oriented grain families showed that Young's modulus $E_{hkl}$ determines the relative order of the lattice strains for both load senses at all temperatures. γ → ε leads to a pronounced TCA of γ lattice strains, where strain accumulation on "soft" austenite grain families is strongly retarded in compression relative to tension. Responsible for this behaviour are the anisotropic driving force and the volume contraction linked to ε formation. α' lattice strains are mostly unaffected by load sense and quickly become larger than γ strains. Consequently, α' quickly carries a higher stress than γ, leading to high macroscopic work hardening, characterising this steel.

Analysis of stress sharing between austenite and martensites revealed another TCA. Elasto-plastic accommodation of the volume expansion linked to α' formation makes γ harden more in compression than in tension. As a result, γ contributes a larger portion of the total stress in compression than in tension. Finally, despite its small volume fraction reaching not more than 5%, ε-martensite can add significant contributions to the macroscopic stress.

## 9. Appendix

### 9.1. Crystallography of γ → α' and γ → ε martensitic transformations

MTs are diffusionless, shape strain dominated structural transformations between parent austenite and product martensite phases [94]. On the continuum scale, the MT is governed by the formation of a habit plane (normal vector ***n***), which becomes the interface between austenite and martensite. ***n*** is an invariant, i.e., undistorted and unrotated plane of austenite. The displacement vector ***d***, describing the deformation on the transforming side of the habit plane (Figure 12), consists of two components: the shear ***s*** parallel to the habit plane and the dilatation ***v*** perpendicular to it. Table 6 lists |***s***| and |***v***| for γ → α' and γ → ε. Values for γ → α' result from the PTMC based on the lattice parameters of γ and α' measured in-situ. |***s***| for γ → ε is the shear needed to obtain an hcp lattice from an fcc lattice by 1/6<112>$_{\gamma}$ slip on alternating $\{111\}_{\gamma}$ planes [95]. The volume contraction for γ → ε was calculated from the in-situ plane spacings of γ and ε along <111>$_{\gamma}$ || $[0001]_{\varepsilon}$.

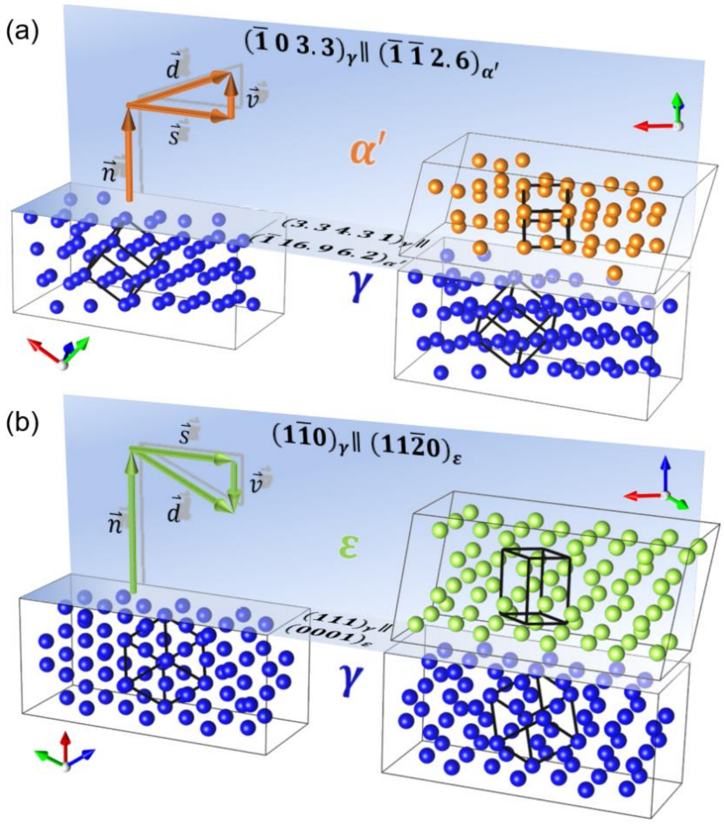

Figure 12: Invariant plane strain of (a) $\gamma \rightarrow \alpha$' and of (b) $\gamma \rightarrow \varepsilon$ for one representative habit plane variant $\boldsymbol{n}$ for each transformation together with the shear $\boldsymbol{s}$ and dilatation $\boldsymbol{v}$ components.

Table 6: Shear and dilatation components of $\gamma \rightarrow \alpha$' and of $\gamma \rightarrow \varepsilon$ in EN 1.4318 (AISI 301LN) steel at 24°C and 60°C. The sign of $|\boldsymbol{v}|$ indicates whether the dilatation causes volume expansion (+) or volume contraction (-), respectively.

| | $\gamma \rightarrow \alpha$' | | $\gamma \rightarrow \varepsilon$ | |
|---|---|---|---|---|
| T (°C) | Shear $|\boldsymbol{s}|$ ($10^{-2}$) | Dilatation $|\boldsymbol{v}|$ ($10^{-2}$) | Shear $|\boldsymbol{s}|$ ($10^{-2}$) | Dilatation $|\boldsymbol{v}|$ ($10^{-2}$) |
| 24 | 22.49 ± 0.05 | (+) 2.4 ± 0.2 | 35.4 | (-) 1.25 ± 0.02 |
| 60 | | | | (-) 1.0 ± 0.1 |

### 9.2. Mechanical driving force *U* for $\gamma \rightarrow \alpha$' and for $\gamma \rightarrow \varepsilon$

$U$ corresponds to the work done by the applied stress $\boldsymbol{\sigma}$ by interacting with the transformation strain. For a habit plane variant (HPV) given by displacement vector $\boldsymbol{d}$ and habit plane normal $\boldsymbol{n}$ (Figure 12), the driving force is [96, 97]

$$U = \boldsymbol{\sigma}\colon \left[\frac{1}{2}(\boldsymbol{n} \otimes \boldsymbol{d} + \boldsymbol{d} \otimes \boldsymbol{n})\right]. \tag{2}$$

Equation 2 assumes that the MT is mainly controlled by the applied stress rather than strain, which is the case for MASS [96, 98]. Looking at one possible HPV in an isolated austenite grain, $U$ depends on the relative orientations of the grain and the applied stress. To this end, we pick one HPV. Rotation of the load axis for the uniaxial tensile stress

$$\boldsymbol{\sigma}_{[100]} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

into all possible crystallographically equivalent directions $<uvw>_\gamma$ of families $<100>_\gamma$, $<110>_\gamma$, and $<111>_\gamma$ is achieved by $\boldsymbol{\sigma}_{<uvw>} = \boldsymbol{R}\boldsymbol{\sigma}_{[100]}\boldsymbol{R}^T$ [99]. $\boldsymbol{R}$ is any (of the infinitely many) rotation matrices aligning the stress axis with <uvw>. To obtain the driving force due to only shear or dilatation (Figure 12), $\boldsymbol{d}$ is replaced by $\boldsymbol{s}$ or $\boldsymbol{v}$, respectively.

### 9.3. Lattice strain extraction from 2D X-ray diffraction ring patterns

The normal component of strain (and similarly of stress) on plane {hkl} is described by

$$\varepsilon_{hkl}(\eta) = \varepsilon^{I}_{hkl} + \varepsilon^{II}_{hkl}(\eta) + \varepsilon^{III}_{hkl}(\eta)$$

where $\eta$ indicates the azimuth in a 2D measurement (see Figure 2). The summed terms on the right hand side are strain types I, II and III, in the spirit of the common phenomenological classification of strain (stress) into macroscopic, intergranular and microscopic contributions [100, 101]. At constant X-ray wavelength, the diffraction angle $2\theta$ for a selected reflection {hkl} is determined by the lattice plane spacing $d_{hkl}$. Variations in $d_{hkl}$ due to type I and II strains give rise to shifts away from the strain-free $2\theta$ positions. Lattice plane spacings thus represent precise, internal strain gauges [102, 103].

By using centred 2D diffraction patterns and by representing Debye-Scherrer rings in polar space ($2\theta$, $\eta$) instead of in pixel coordinates (as in Figure 4) hydrostatic and deviatoric type I lattice strains can be separated from each other in a straightforward manner. Figure 13 illustrates the influence of both strain (stress) types on 2D diffraction patterns in polar space.

Hydrostatic type I strain, corresponding to the average strain across all diffracting grains and lattice planes, manifests itself as a shift to higher or lower diffraction angles for compression and tension, respectively, independent of the azimuth $\eta$. Deviatoric type I strain in contrast represents a deviation from the hydrostatic average, leading to a characteristic sinusoidal variation of $2\theta$ with $\eta$. In the absence of deviatoric type I strain each reflection is a straight line at constant diffraction angle $2\theta$.

Conversely, if deviatoric type I strains are present, the diffraction angles exhibit periodic variations that repeat every $\Delta\eta$ = 180°. The superposition of both strains gives rise to sinusoidal variations shifted away from the strain-free reference configuration (Figure 13). Type III strain (stress) leads to broadening of the reflection along $2\theta$. Such strain is accessible by peak profile analysis and is not considered in the present context.

For uniaxial deformation the position of the minimum (maximum) $2\theta$ values along $\eta$ is directly related to the load axis orientation on the detector. In the present work, the load axis aligns with $\eta$ = 0° (Figure 2). Therefore, minimum, and maximum $2\theta$ angles are located at $\eta$ = 0°, 180°, 360° for tension and compression, respectively.

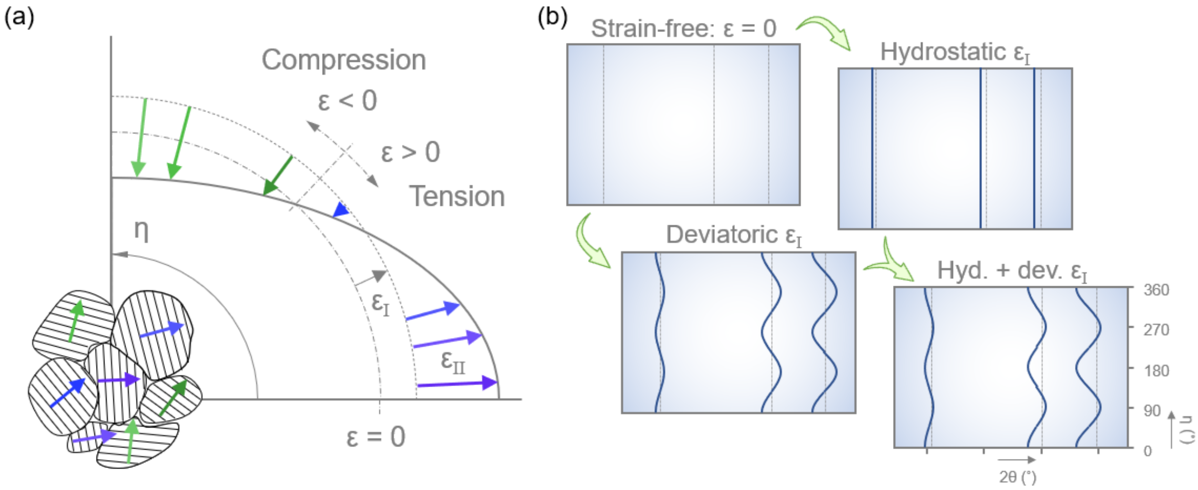


Figure 13: (a) Illustration of hydrostatic and deviatoric type I strains depending on azimuth $\eta$ and (b) their influence on 2D diffraction patterns in polar space. Both (a) and (b) assume tensile deformation along the horizontal direction ($\eta$ = 0°).

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**Author contributions**

**Competing interests**

**Data availability**